

Exercise 4

Hugo Lladró Prats, Inés Mang Román, Ahmet Meriç Kızıltaş

As usual we will use pandas and seaborn for the creation of graphics. We begin by installing the libraries.

```
In [1]: !mamba install pandas
        !mamba install seaborn
```

mambajs 0.19.13

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, pandas

Channels: emscripten-forge, conda-forge

Solving environment...

Solving took 0.40079999999701976 seconds

Name Channel	Version	Build
+ pandas emscripten-forge	3.0.2	np22py313h9d9dc1e_0
+ python-tzdata conda-forge	2026.1	pyhd8ed1ab_0
- pip conda-forge	25.3	pyh145f28c_0
mambajs 0.19.13		

Specs: xeus-python, numpy, matplotlib, pillow, ipywidgets>=8.1.6, ipyleaflet, sci
py, pandas, seaborn

Channels: emscripten-forge, conda-forge

Solving environment...

Solving took 0.40920000000298024 seconds

Name Channel	Version	Build
+ patsy emscripten-forge	1.0.2	py313h1804a44_1
+ seaborn conda-forge	0.13.2	hd8ed1ab_3
+ seaborn-base conda-forge	0.13.2	pyhd8ed1ab_3
+ statsmodels emscripten-forge	0.14.6	np22py313h3cf259a_1

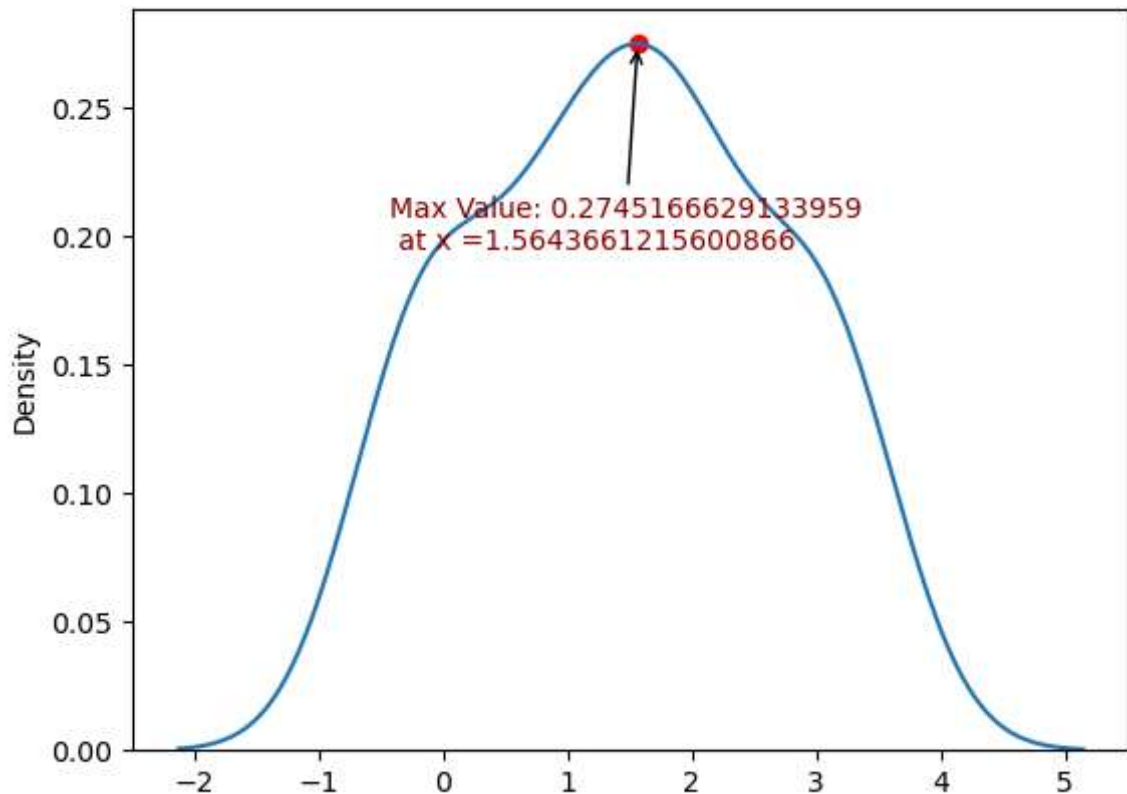
```
In [2]: import numpy as np
        import pandas as pd
        import seaborn as sns
        import matplotlib.pyplot as plt
        import scipy as sp
```

a) The file values.txt that is available on eCampus contains 40 scalar values. Use the seaborn library with default parameters to visualize the distribution of those values using

Kernel Density Estimation. Based on the visualization, at which location(s) do you think the underlying distribution has a mode (peak)? (2P)

```
In [19]: data = pd.read_csv('values.txt', sep=" ", header=None)

plot = sns.kdeplot(data=data, legend = False)
x, y = plot.lines[0].get_data()
plt.scatter(x[np.argmax(y)], y.max(), color = 'r', label = 'maximum')
plt.annotate(f'Max Value: {y.max()}\n at x={x[np.argmax(y)]}',
            xy=(x[np.argmax(y)], y.max()),
            xytext=(x[np.argmax(y)]-2, y.max()-0.08),
            arrowprops=dict(facecolor='black', arrowstyle='->'),
            fontsize=10, color='darkred')
plt.show()
```



Based on the figure, we observe that the plot seems to attain its maximum at $x = 1.56$, at 0.27. The distribution is unimodal and roughly normal; the area around the mode shows a peak, while the value quickly drops outside of $[0, 3]$.

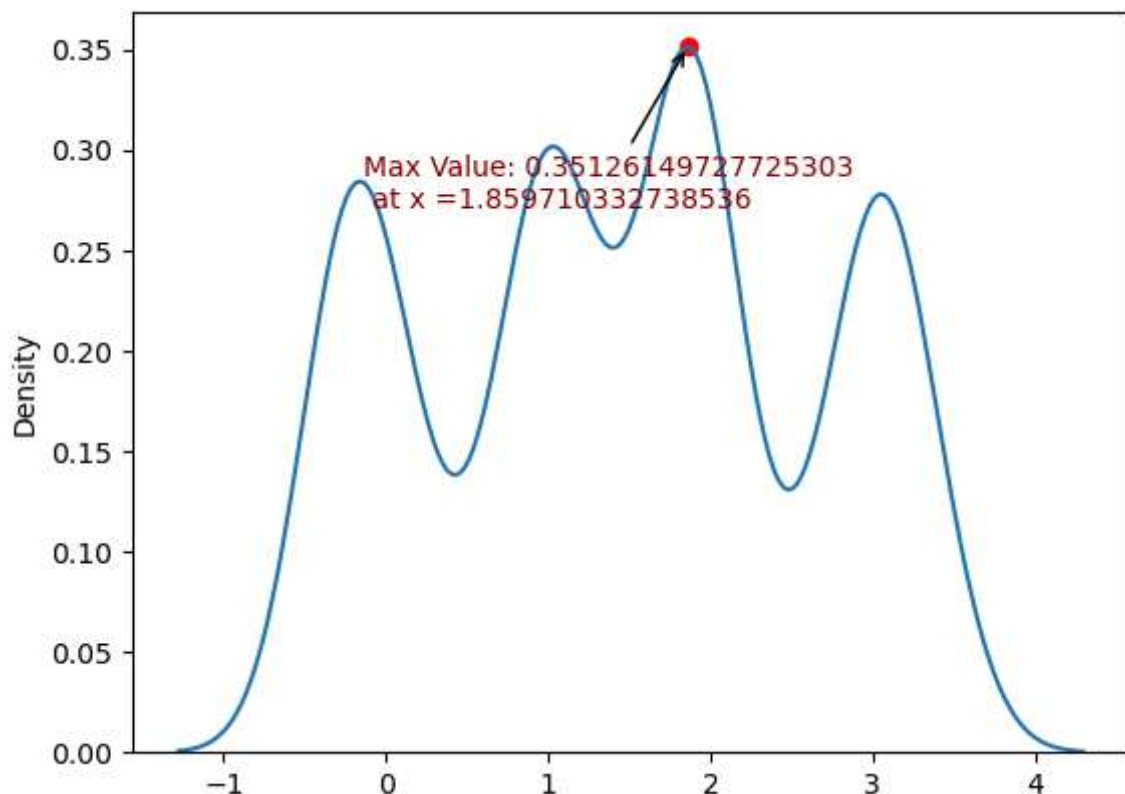
b) Produce alternative KDE plots by adjusting the bandwidth to higher and lower values. Briefly describe in your own words how this changes the shape of the estimated distribution. Visualize the same data in a different way to help you decide which setting most faithfully reflects the distribution which generated the data. In particular, at which location(s) would you assume it has modes? (3P)

```
In [20]: plot = sns.kdeplot(data=data, legend = False, bw_adjust = 0.5)
x, y = plot.lines[0].get_data()
plt.scatter(x[np.argmax(y)], y.max(), color = 'r', label = 'maximum')
plt.annotate(f'Max Value: {y.max()}\n at x={x[np.argmax(y)]}',
            xy=(x[np.argmax(y)], y.max()),
            xytext=(x[np.argmax(y)]-2, y.max()-0.08),
```

```

arrowprops=dict(facecolor='black', arrowstyle='->'),
                fontsize=10, color='darkred')
plt.show()

```

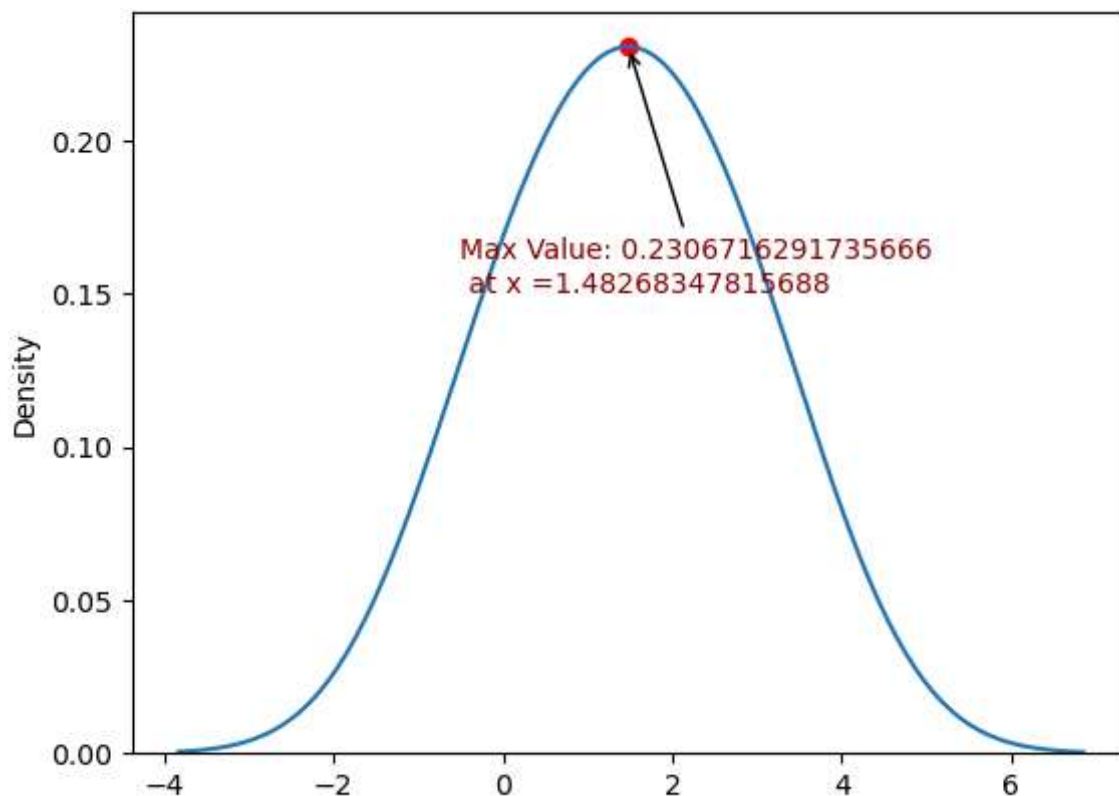


If we use half the standard bandwidth, we now see that the distribution has several peaks, while the mode is at $x = 1.85$. The shape is also qualitatively different, no longer unimodal, and with no semblance of normality.

```

In [22]: plot = sns.kdeplot(data=data, legend = False, bw_adjust = 2)
x, y = plot.lines[0].get_data()
plt.scatter(x[np.argmax(y)], y.max(), color = 'r', label = 'maximum')
plt.annotate(f'Max Value: {y.max()}\n at x={x[np.argmax(y)]}',
            xy=(x[np.argmax(y)], y.max()),
            xytext=(x[np.argmax(y)]-2, y.max()-0.08),
            arrowprops=dict(facecolor='black', arrowstyle='->'),
            fontsize=10, color='darkred')
plt.show()

```

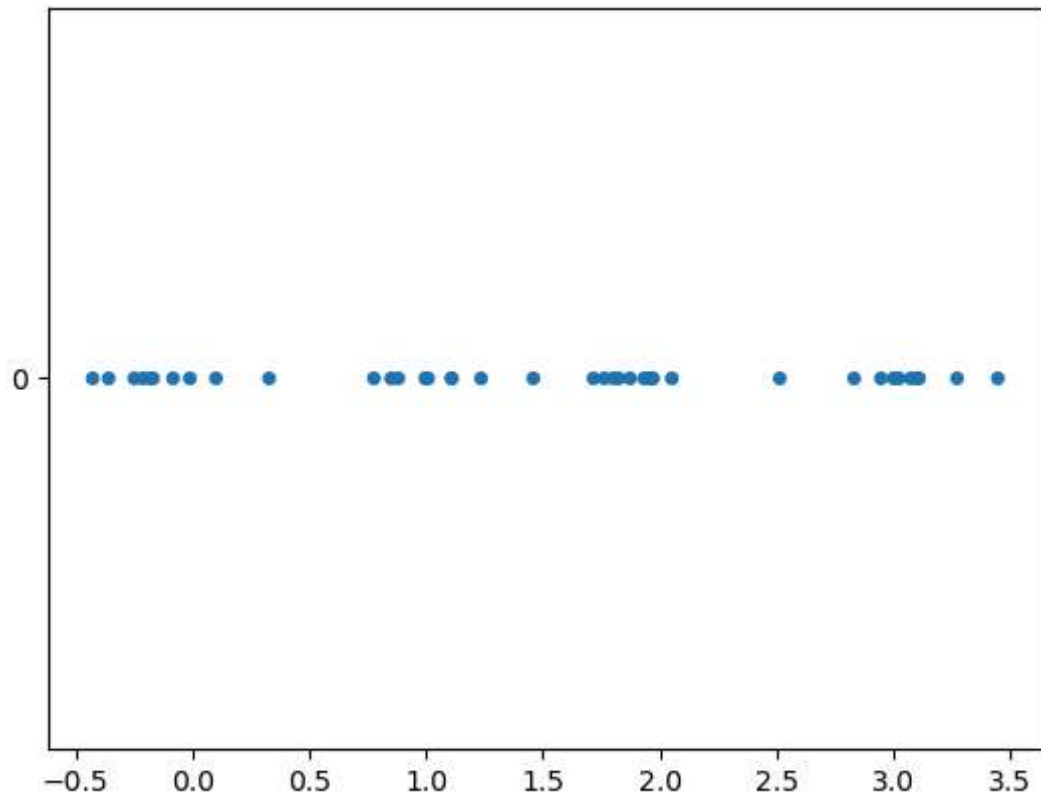


If we use double the standard bandwidth, the distribution now looks almost like a normal distribution centered at the mode $x = 1.48$. We cannot observe the detail of the individual observations.

Now we will use a strip plot to observe the actual observations and try to detect the underlying pattern.

```
In [28]: sns.stripplot(data=data, orient = 'h', jitter = False)
```

```
Out[28]: <Axes: >
```



In this visualization we observe that there are roughly 4 clusters of data, centered at $x = -0.25, 1, 1.75$ and 3 respectively. The greatest concentration of observations sits around $x = 1.75$ so the mode should be there.